

05 — DNS (Domain Name System)

Introduction

In the previous topic, I learned about **domain names** and how they provide human-friendly names for Internet services.

The next question is:

How does a computer turn a domain name like `example.com` into information it can use to reach the correct server?

The answer involves **DNS — the Domain Name System**.

DNS is one of the fundamental systems that makes the Internet easier to use.

Without DNS, users would often need to remember IP addresses instead of domain names.

What Is DNS?

DNS stands for:

Domain Name System

DNS is a distributed naming system used to translate domain names

into information such as IP addresses.

For example:

```
example.com
  ↓
  DNS
  ↓
93.184.216.34
```

The browser can then use the resulting network information to communicate with the appropriate server.

A simplified idea is:

```
Human
  ↓
Domain Name
  ↓
DNS
  ↓
IP Address
  ↓
Server
```

📁 Why Do We Need DNS?

Computers communicate across networks using addresses such as IP addresses.

For example:

```
93.184.216.34
```

But remembering IP addresses for every website would be difficult.

Imagine trying to remember:

```
Website A → 192.0.2.10  
Website B → 192.0.2.20  
Website C → 192.0.2.30  
Website D → 192.0.2.40
```

Instead, we can use:

```
website-a.com  
website-b.com  
website-c.com  
website-d.com
```

DNS provides the system that allows these human-readable names to be resolved.

DNS Is Like an Internet Directory

A simple way I understand DNS is to think of it as a **directory for Internet names**.

If I know:

```
example.com
```

DNS helps me find the information associated with that name.

Similar to looking up someone's name in a contact list:

Name



Contact Information

DNS can be thought of as:

Domain Name



DNS Information



IP Address / Other Records

However, DNS is much more powerful than a simple phonebook because it is a distributed system containing different types of records and servers.

4 DNS and the Web Browser

When I enter:

```
https://example.com
```

into a browser, the browser needs to determine where to send the request.

A simplified process is:

Browser



"What is the IP address for example.com?"

DNS Resolver



DNS System



IP Address



Browser

The browser can then continue connecting to the server.

5📖 What Is a DNS Resolver?

A **DNS resolver** is responsible for finding the DNS information requested by a client.

A client usually does not directly communicate with every DNS server involved in the process.

Instead, it commonly sends a query to a **recursive DNS resolver**.

For example:

Browser



Operating System



DNS Resolver



DNS Hierarchy

The resolver does the work of finding the answer and returning it to the client.

6 Recursive DNS Resolver

A recursive resolver receives a DNS query and attempts to obtain the requested answer.

For example:



The resolver may already have the answer in its cache.

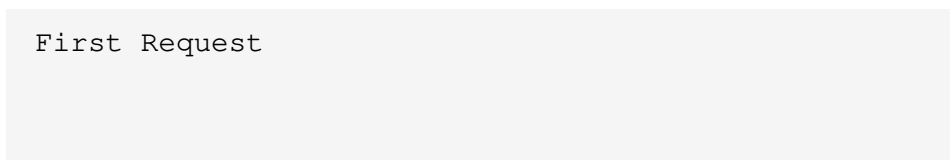
If it does not, it may need to query other DNS servers.

7 DNS Caching

DNS queries are often cached.

Caching means temporarily storing information so it can be reused later.

For example:



```
Client
  ↓
Resolver
  ↓
DNS lookup
  ↓
IP Address
```

Second Request

```
Client
  ↓
Resolver
  ↓
Cached Answer
  ↓
IP Address
```

Caching can reduce:

- Lookup time
- Network traffic
- DNS server workload

TTL — Time To Live

DNS records can have a **TTL**, or Time To Live.

TTL tells DNS systems how long a record can be cached before it should be considered expired and queried again.

For example:

```
DNS Record
```

TTL = 3600 seconds

This means the record can generally be cached for:

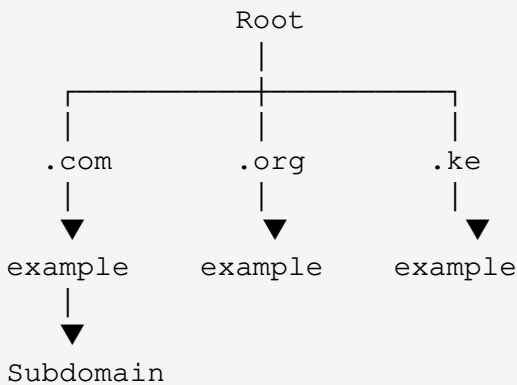
3600 seconds
= 1 hour

TTL values depend on how the domain's DNS is configured.

📖 The DNS Hierarchy

DNS is organized hierarchically.

A simplified structure is:



The major levels include:

1. Root
 2. Top-Level Domain (TLD)
 3. Authoritative domain servers
-

Root DNS Servers

At the top of the DNS hierarchy is the **DNS root**.

The root system directs queries toward the appropriate TLD name servers.

For example, if the query concerns:

```
example.com
```

the root system helps direct the resolver toward the `.com` TLD infrastructure.

The root does not normally provide the final IP address for every website.

Instead, it helps guide the query.

TLD Name Servers

TLD stands for:

Top-Level Domain

Examples include:

```
.com  
.org  
.net  
.ke  
.uk  
.id
```

TLD name servers know which authoritative name servers are responsible for domains within their TLD.

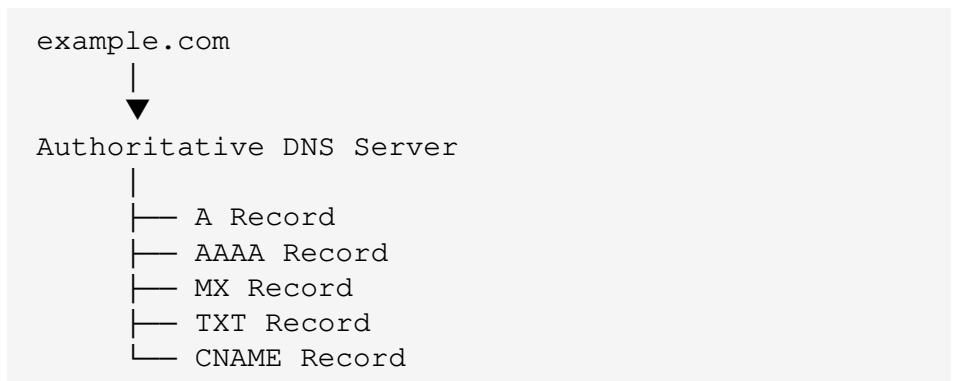
For example:



1.2 Authoritative DNS Servers

An **authoritative DNS server** contains the authoritative DNS records for a domain.

For example:



The authoritative server is ultimately responsible for providing the DNS information for that domain.

13 How a DNS Lookup Works

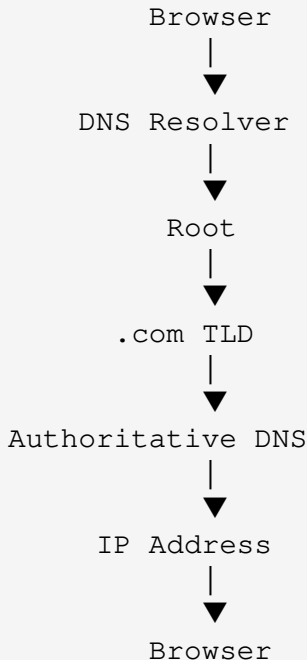
Let's look at a simplified example.

I enter:

```
example.com
```

into my browser.

The process may look like:



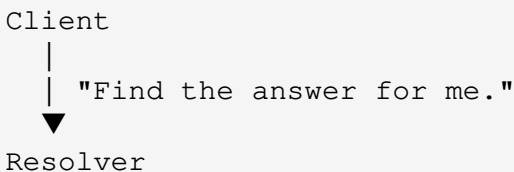
The resolver then returns the answer to the client.

14 Recursive vs Iterative Queries

There are two concepts I need to understand.

Recursive Query

The client asks the resolver to find the answer.

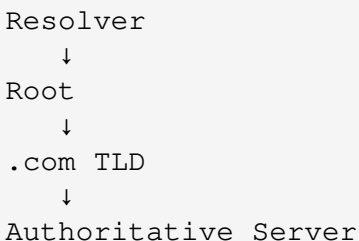


The resolver does the work.

Iterative Process

The resolver may ask DNS servers for directions.

For example:



Each server can provide information that helps the resolver continue.

15 DNS Records

DNS contains different types of records.

Some important ones are:

Record	Purpose
A	Maps a name to an IPv4 address
AAAA	Maps a name to an IPv6 address
CNAME	Creates an alias to another domain name
MX	Specifies mail servers
NS	Identifies authoritative name servers
TXT	Stores text information
PTR	Used for reverse DNS lookups
SOA	Contains information about a DNS zone

16 A Record

An **A record** maps a domain name to an IPv4 address.

Example:

```
example.com
  ↓
```

A



192.0.2.10

Conceptually:

```
example.com → 192.0.2.10
```

AAAA Record

An **AAAA record** is used to map a domain name to an IPv6 address.

Example:

```
example.com
```



```
AAAA
```



```
2001:db8::10
```

The difference is:

```
A      → IPv4
```

```
AAAA   → IPv6
```

CNAME Record

A **CNAME** record creates an alias from one domain name to another domain name.

For example:

```
www.example.com
```



```
CNAME
```



```
example.com
```

This means `www.example.com` can be configured as an alias for `example.com`.

100 MX Record

An **MX record** specifies the mail servers responsible for receiving email for a domain.

For example:

```
example.com
```



```
MX
```



```
mail.example.com
```

This is important when setting up email services for a domain.

200 NS Record

An **NS record** identifies the name servers that are authoritative for a DNS zone.

For example:

```
example.com
  ↓
NS
  ↓
ns1.example-dns.com
ns2.example-dns.com
```

These servers can contain the authoritative DNS records for the domain.

2.1.1 TXT Record

A **TXT record** stores text information associated with a domain.

TXT records are commonly used for things such as:

- Domain verification
- Email authentication
- SPF information
- Other service configurations

For example:

```
example.com
  ↓
TXT
  ↓
"verification=value"
```

2.2.1 Reverse DNS

Normal DNS usually works like:

```
Domain
↓
IP Address
```

Reverse DNS works in the opposite direction:

```
IP Address
↓
Domain Name
```

This commonly involves a **PTR record**.

For example:

```
192.0.2.10
↓
PTR
↓
example.com
```

Reverse DNS is useful for various networking and email-related purposes.

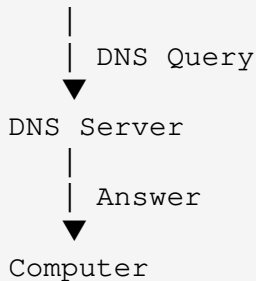
2.3 DNS and Networking

DNS is not only important for web development.

It is also a fundamental networking service.

For example, on a network:

```
Computer
```



This is something I can encounter when working with:

- Routers
- DHCP
- Servers
- Firewalls
- Active Directory
- Network troubleshooting
- Internet services

24 DNS Troubleshooting

DNS problems can cause websites and services to become unreachable even when the server itself is working.

For example:

```
graph TD; A[User] --> B[Website]; B -- "X" --> C[DNS Resolution Failed];
```

Some useful commands for investigating DNS include:

Windows

```
nslookup example.com
```

Linux

```
dig example.com
```

or:

```
nslookup example.com
```

These tools can show information about DNS resolution.

2 DNS Example

Imagine I create:

```
mukhtaartech.com
```

I want the website to point to my web server.

I could configure an A record such as:

```
mukhtaartech.com
    ↓
A
    ↓
[Web Server IPv4 Address]
```

Then I might create:

```
www.mukhtaartech.com
```

as an alias:

```
www.mukhtaartech.com
  ↓
  CNAME
  ↓
mukhtaartech.com
```

For email, I could have:

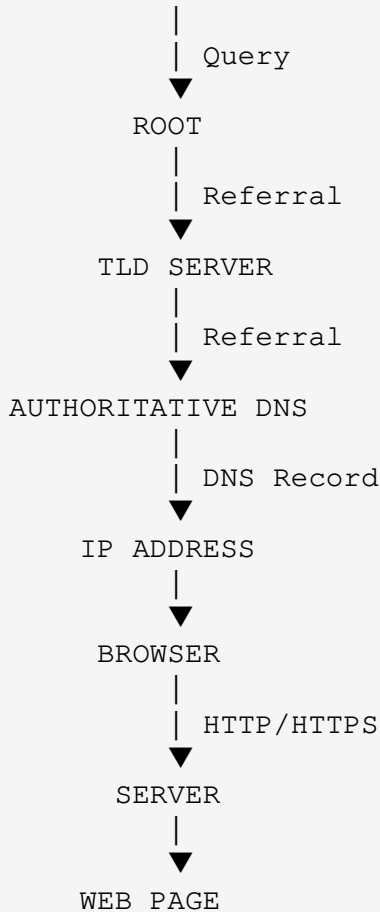
```
mukhtaartech.com
  ↓
  MX
  ↓
Mail Server
```

This demonstrates how one domain can use different DNS records for different services.

Complete DNS Journey

Putting everything together:

```
USER
  |
  ▼
BROWSER
  |
  | Domain
  ▼
DNS RESOLVER
```



What I Learned

From this topic, I learned that DNS is a distributed system that allows humans to use domain names while computers and network services can use information such as IP addresses.

I learned that:

- DNS stands for Domain Name System.
 - DNS is hierarchical.
 - DNS resolvers help clients find DNS information.
 - DNS information can be cached.
 - TTL controls how long DNS information can generally remain cached.
 - Root DNS helps direct queries toward TLD infrastructure.
 - TLD servers help identify authoritative DNS servers.
 - Authoritative DNS servers provide the actual DNS records for a domain.
 - Different DNS records serve different purposes.
 - DNS is important for both web development and networking.
 - DNS troubleshooting is an important networking skill.
-



Key Terms

Term	Meaning
DNS	Domain Name System
DNS Resolver	System that finds DNS information for clients
Recursive Resolver	Resolver that performs the lookup process on behalf of a client
Root	Top level of the DNS hierarchy
TLD	Top-Level Domain
Authoritative DNS	DNS server responsible for a domain's authoritative records
A Record	Maps a name to an IPv4 address
AAAA Record	Maps a name to an IPv6 address
CNAME	Alias for another domain name
MX	Specifies mail servers

NS	Identifies authoritative name servers
TXT	Stores text information
PTR	Used for reverse DNS
TTL	Time To Live; controls DNS caching duration

Practical Experiment

I want to test DNS resolution myself.

Windows

Open Command Prompt and run:

```
nslookup google.com
```

Then try:

```
nslookup github.com
```

I can observe:

- The DNS server being used
- The domain being queried
- The IP addresses returned

I can also try:

```
nslookup -type=MX google.com
```

to investigate mail-related DNS records.

? Questions I Want to Explore Next

- What exactly happens during DNS propagation?
 - Why can DNS changes take time to appear?
 - What is DNS caching at different levels?
 - What is a DNS zone?
 - What is a nameserver?
 - What is the difference between a DNS resolver and a nameserver?
 - How does DNS over HTTPS work?
 - How does DNS over TLS work?
 - How can I configure DNS for my own domain?
-

Progress

Stage: Web Fundamentals

Topic: DNS (Domain Name System)

Status: ● Learning

Previous: [Domain Names](#)

Next: Web Hosting

DNS is one of the systems that turns human-friendly names into information that allows Internet services to be found.